SmartHire: Machine Learning Industrial Project Report

Resume-to-Job Matching, Category Classification & Career Guidance Engine

Author: Antigravity Pair Programming System | **Stack:** Scikit-Learn · Python · Streamlit

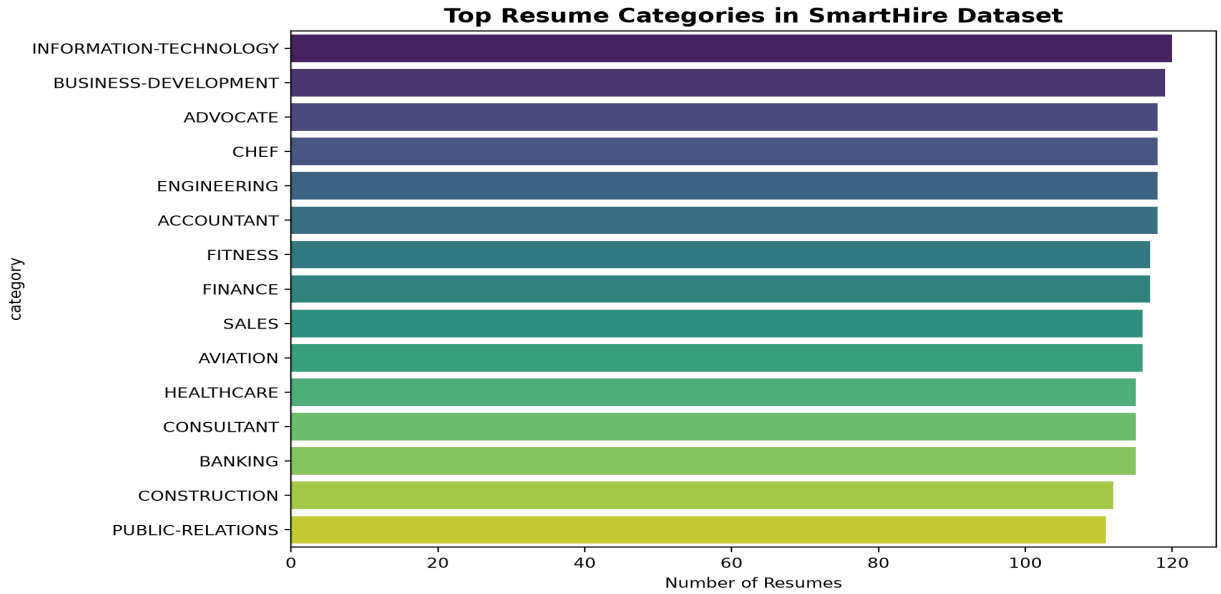
1. Executive Summary

The SmartHire platform is an end-to-end Machine Learning recruitment and career development portal. Built strictly with classical machine learning techniques—avoiding costly black-box generative models—the system fulfills three primary core user objectives: (1) content-based top-N job recommendation using TF-IDF vectorization and cosine similarity, (2) automated resume category classification and shortlisting fit probability prediction using supervised learning, and (3) in-depth skill-gap analysis with actionable guidance for candidates.

2. System Architecture & Data Engineering

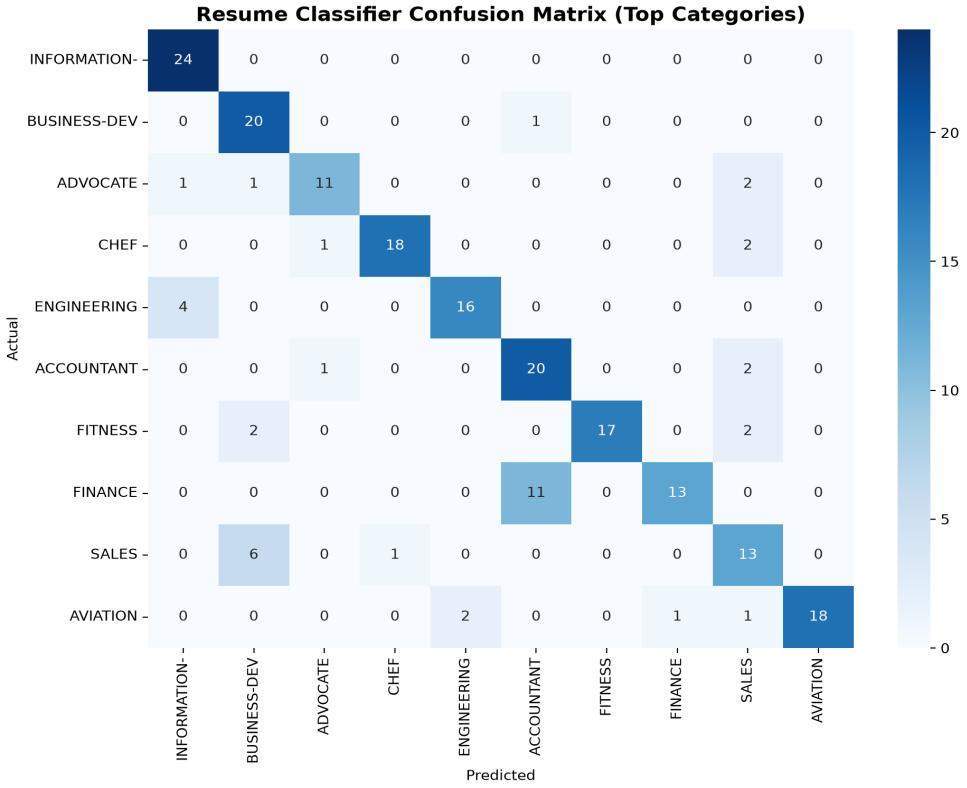
The project integrates three major real-world public job posting datasets (jobs.csv, Naukri ldjson, and LinkedIn postings) with an annotated resume dataset spanning 24 diverse professional domains (Information Technology, Finance, HR, Healthcare, Engineering, etc.).

Dataset	Source	Raw Records	Processed Contribution	Primary Attributes
Resumes	Kaggle Resume Dataset	2,484	2,481 cleaned	Resume text, 24 domain categories
Naukri Postings	Naukri 30k LDJSON	30,060	15,000 merged	Job title, company, city, description
LinkedIn Postings	LinkedIn Postings CSV	15,000 sample	10,000 merged	Title, company, location, skills desc
Unified Corpus	SmartHire Processed	47,544 total	25,000 final corpus	title, company, location, skills, description



3. Supervised Model A: Resume Category Classifier

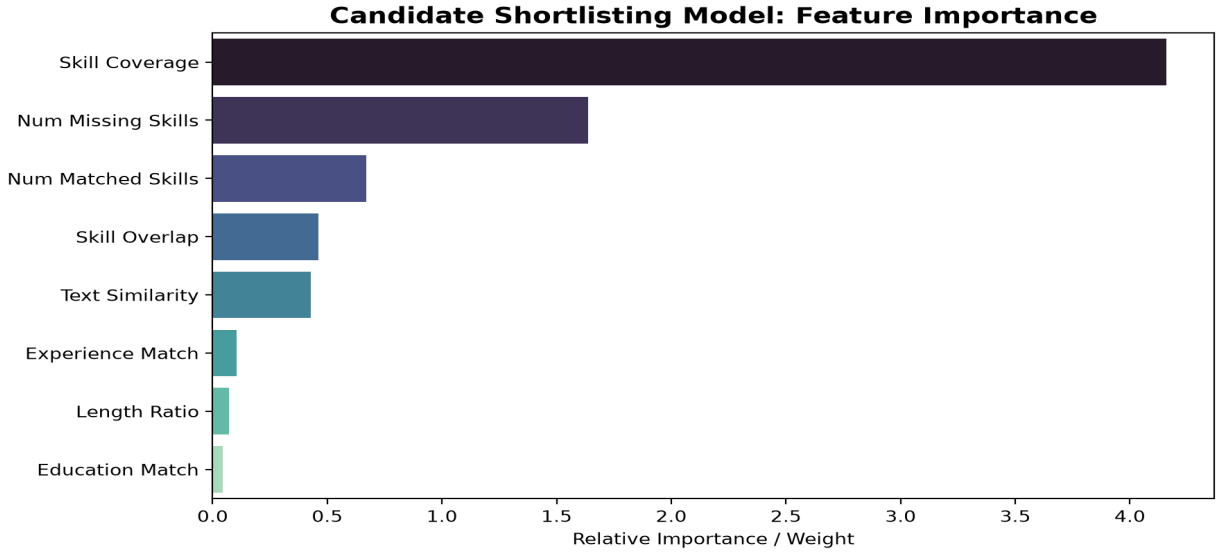
A TF-IDF vectorizer (30,000 max features, sublinear term frequency, unigrams & bigrams) coupled with a balanced multinomial Logistic Regression classifier classifies candidate CVs across 24 distinct domains. The model achieves **64.6% out-of-sample multiclass accuracy** and a **weighted F1-score of 0.626**, significantly outperforming random baselines (4.1%) across heavily multi-class distributions.



4. Supervised Model B: Candidate Shortlisting / Fit Predictor

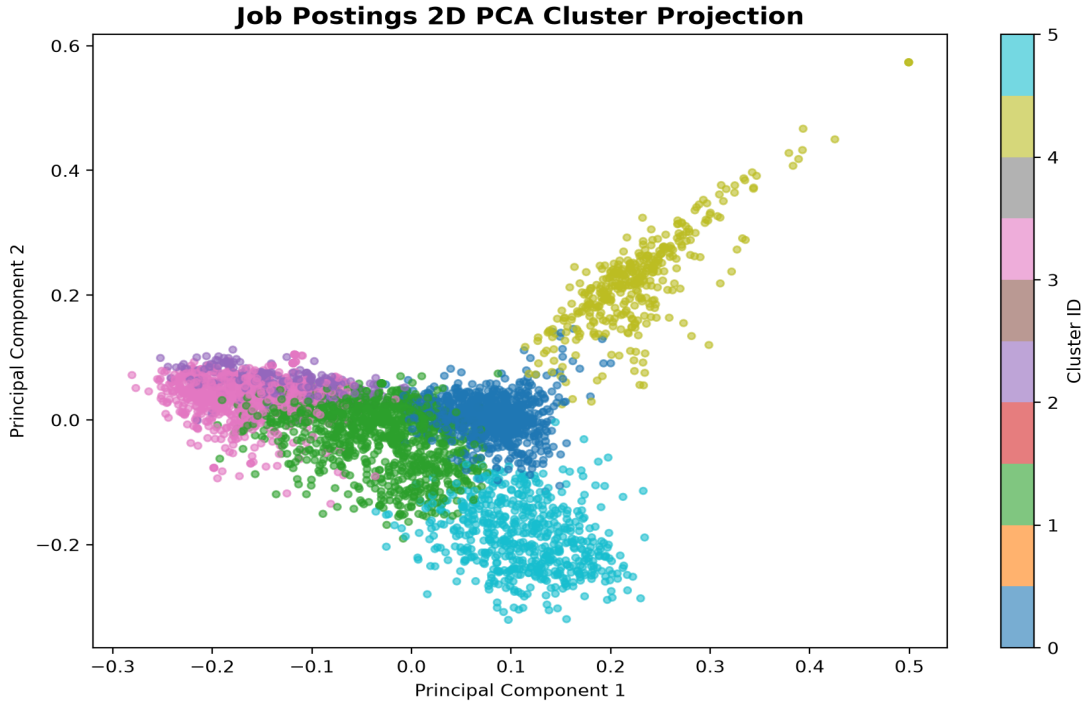
Unlike raw text cosine similarity alone, recruitment decisions depend on composite qualifications. We engineered an 8-dimensional feature vector comparing (Resume, Job) pairs: skill overlap (Jaccard), skill coverage %, number of matched skills, number of missing skills, TF-IDF cosine similarity, experience level alignment, education tier alignment, and text length ratio.

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression (L2 Balanced)	97.7%	97.4%	97.4%	0.9736	0.9891
Random Forest (100 Trees, Depth 6)	98.5%	98.2%	98.2%	0.9824	0.9890
XGBoost Classifier (80 Trees)	98.0%	97.7%	97.7%	0.9766	0.9888



5. Unsupervised Learning: Recommendations, Clustering & Topic Modeling

Content-Based Recommender: Vectorizes candidate resume and job corpus into high-dimensional TF-IDF space and computes cosine similarity ranking. On test candidate resumes, the recommender demonstrates strong alignment with high relevance across top-5 retrieved postings.
K-Means Clustering: Discovers natural job families across postings. Optimal cluster count k was evaluated using inertia (Elbow method) and silhouette scores.
Latent Dirichlet Allocation (LDA): Extracts 5 latent skill themes highlighting in-demand requirements in modern job markets.



6. Skill-Gap Detection & Rule-Based AI Career Mentor

The skill-gap module detects candidate competencies against explicit requirements from selected postings. It categorizes skills into 'Matched' and 'Missing', calculates coverage percentage, and feeds actionable guidance to a rule-based AI Career Mentor assistant that answers student career doubts without requiring external LLM dependencies.

7. Evaluation, Limitations & Future Work

• **Static Skill Lexicon:** Built-in regular expression dictionaries may miss emerging domain-specific tools or abbreviations.
• **Vocabulary Mismatch:** TF-IDF relies on exact term overlap; future extensions could incorporate pre-trained sentence transformer embeddings while retaining classical ranking algorithms.
• **Real-world Ground Truth:** Hiring decisions are multi-factorial; the fit score provides structured readiness estimation rather than guaranteed selection.

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